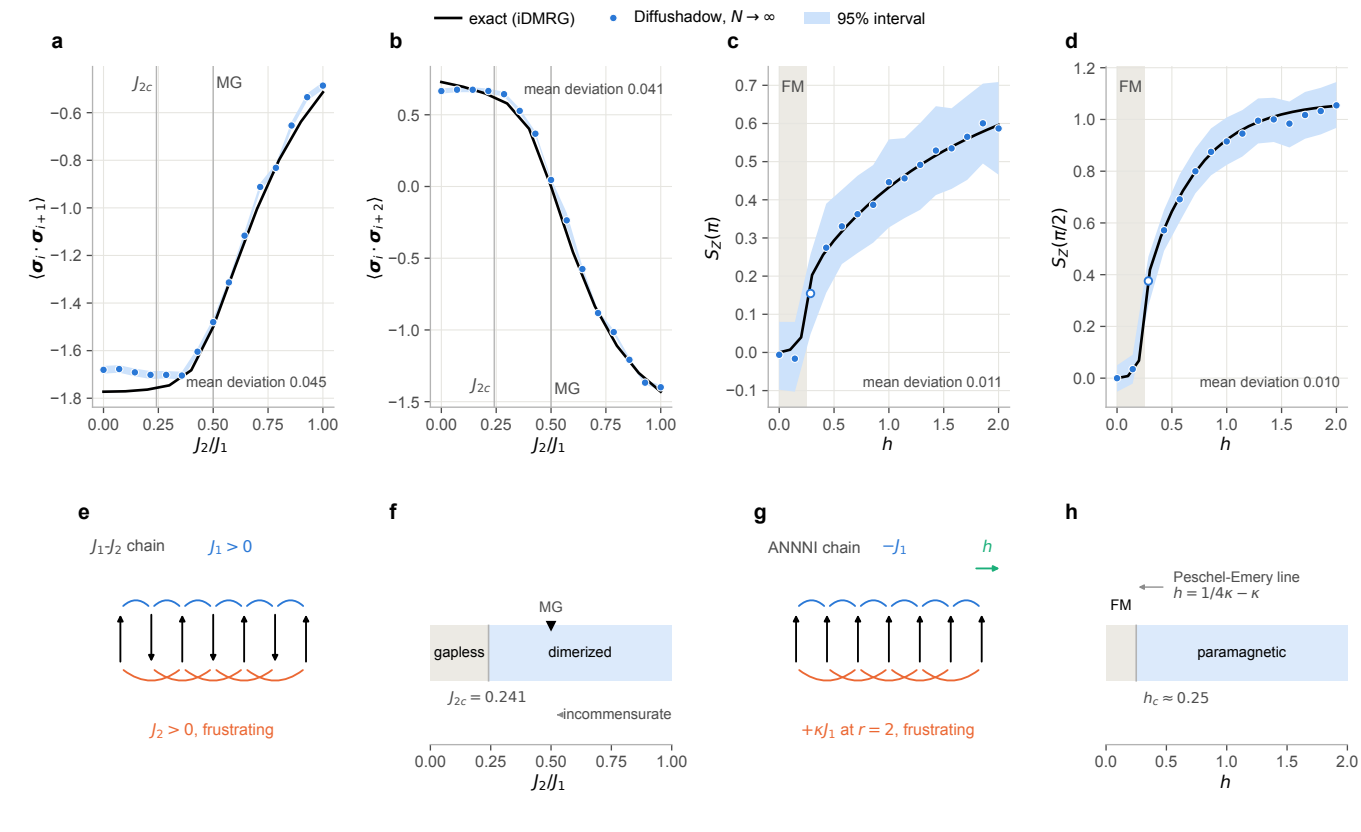

exact (iDMRG)
Diffushadow, $N \to \infty$
95% interval
a
J_{2c}
MG
mean deviation 0.045
$\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+1} \rangle$
J_2/J_1
b
mean deviation 0.041
J_{2c}
MG
$\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+2} \rangle$
J_2/J_1
c
FM
mean deviation 0.011
$S_z(\pi)$
h
d
FM
mean deviation 0.010
$S_z(\pi/2)$
h
e
J_1-J_2 chain
$J_1 > 0$
$J_2 > 0$, frustrating
f
MG
gapless
dimerized
$J_{2c} = 0.241$
incommensurate
J_2/J_1
g
ANNNI chain
$-J_1$
h
$+\kappa J_1$ at $r = 2$, frustrating
h
Peschel-Emery line
$h = 1/4\kappa - \kappa$
FM
paramagnetic
$h_c \approx 0.25$
h